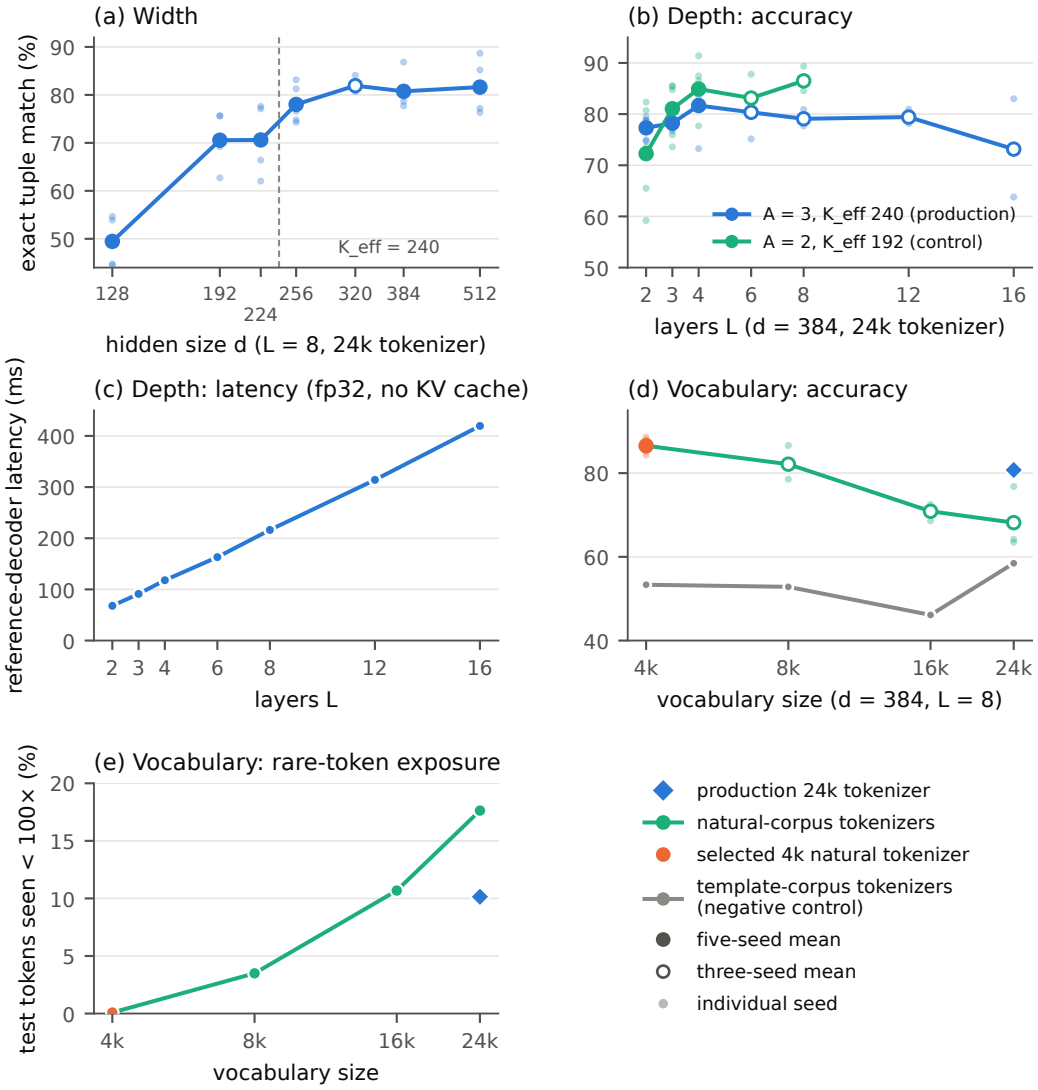
(a) Width
exact tuple match (%)
90
80
70
60
50
K_eff = 240
128
192
224
256
320
384
512
hidden size d (L = 8, 24k tokenizer)
(b) Depth: accuracy
90
80
70
60
50
A = 3, K_eff 240 (production)
A = 2, K_eff 192 (control)
2
3
4
6
8
12
16
layers L (d = 384, 24k tokenizer)
(c) Depth: latency (fp32, no KV cache)
reference-decoder latency (ms)
400
300
200
100
0
2
3
4
6
8
12
16
layers L
(d) Vocabulary: accuracy
80
60
40
4k
8k
16k
24k
vocabulary size (d = 384, L = 8)
(e) Vocabulary: rare-token exposure
test tokens seen < 100× (%)
20
15
10
5
0
4k
8k
16k
24k
vocabulary size
production 24k tokenizer
natural-corpus tokenizers
selected 4k natural tokenizer
template-corpus tokenizers
(negative control)
five-seed mean
three-seed mean
individual seed